Physical Cume
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For an unknown diatomic molecule the following vibrational transitions were observed: for the transition from v=0 to v=1, 284.5 cm^{-1}; for the transition from v=1 to v=2, 283.0 cm^{-1}; and for the transition from v=2 to v=3, 281.5 cm^{-1}. Use the equations for the anharmonic oscillator obtained with the Morse potential energy function to solve the following problems.

1.)Calculate the zero-point energy of this molecule (in cm^{-1}).
2)Calculate the harmonic oscillator frequency, $\omega/2\pi c$, (in cm^{-1}) for this molecule.
3)Calculate the dissociation energy for this molecule (in cm^{-1}).
4.)Calculate the transition from v=3 to v=4 (in cm^{-1}).

HELP:

The harmonic approximation

In regions close to R_e (at the minimum of the curve) the potential energy can be approximated by a parabola, and we can write

$$V = \tfrac{1}{2}k(R - R_e)^2$$

$$E_v = (v + \tfrac{1}{2})\hbar\omega \qquad \omega = \left(\frac{k}{\mu}\right)^{1/2} \qquad v = 0, 1, 2, \ldots$$

The **vibrational terms** of a molecule, the energies of its vibrational states expressed in wavenumbers, are denoted by G, and so

$$G(v) = (v + \tfrac{1}{2})\bar{\nu} \qquad \bar{\nu} = \frac{\omega}{2\pi c}$$

Anharmonicity

Morse potential energy is

$$V = D_e\{1 - e^{-a(R-R_e)}\}^2$$

where D_e is the depth of the potential minimum and

$$a = \left(\frac{\mu}{2D_e}\right)^{1/2}\omega$$

Schrödinger equation can be solved for this potential and the permitted energy levels are

$$G(v) = (v + \tfrac{1}{2})\bar{\nu} - (v + \tfrac{1}{2})^2 x_e\bar{\nu} \qquad x_e = \frac{a^2\hbar}{2\mu\omega} \qquad (24)$$

x_e is called the **anharmonicity constant**.